FrameworkMissionRequirements

«requirement»

<G-04> Complete Contrasting Evaluation Cases

doc

The framework completes both selected contrasting EPS evaluation cases with the common core information model and workflow, while identifying every approved case-specific parameter, extension, and mapping. Evidence basis: buruso2026modelbased Sec. 5.1 and Table 5; vizitiu2025optimizing Sec. 4.7 and Table 3; rostamnia2026digitaltwin Sec. 8.4.

subject

enterprise : 'Academic Nanosatellite EPS Development Enterprise'

require constraints

docBoth contrasting evaluation cases complete with all case-specific adaptations declared.
statement { }

«#Refinement»

«analysis»

Framework Candidate Eligibility

attributes

G-01 satisfied : Boolean
G-02 satisfied : Boolean
G-03 satisfied : Boolean
G-04 satisfied : Boolean
mandatory system requirements satisfied : Boolean
eligible : Boolean

constraints

eligibility : 'Candidate Eligibility Rule' { in 'G-01 satisfied' = 'G-01 satisfied'; in 'G-02 satisfied' = 'G-02 satisfied'; in 'G-03 satisfied' = 'G-03 satisfied'; in 'G-04 satisfied' = 'G-04 satisfied'; in 'mandatory system requirements satisfied' = 'mandatory system requirements satisfied'; out eligible = eligible; }

«requirement»

<G-03> Resolve Critical Evidence Inconsistencies

doc

A framework evaluation baseline has no unresolved discrepancy classified as critical among its requirements, model properties, interfaces, budgets, assumptions, and analysis inputs.
An unavailable, incomplete, incompatible, or untraceable external artifact is rejected without changing controlled content, and its validation discrepancies are retained and returned through the applicable interface.
Evidence basis: vizitiu2025optimizing Sec. 4.7 and Table 3; buruso2026modelbased Sec. 5.1 and Table 5; rostamnia2026digitaltwin Table 13.

subject

enterprise : 'Academic Nanosatellite EPS Development Enterprise'

require constraints

docNo critical evidence inconsistency remains unresolved, and no discrepancy from a rejected artifact is hidden.
statement { }

«#Refinement»

«requirement»

<G-00> Candidate Framework Eligibility

doc

A candidate framework physical architecture enters comparison ranking only after it satisfies G-01 through G-04 and every mandatory system requirement. This eligibility conjunction is the evaluation rule adopted by this study.

subject

enterprise : 'Academic Nanosatellite EPS Development Enterprise'

require constraints

docOnly a feasible candidate enters comparison ranking.
statement { }

«#Refinement»

«concern»

^earlyEPSDesignJustification

subject

^enterprise : 'Academic Nanosatellite EPS Development Enterprise'

stakeholders

^missionTeam : 'Academic Mission Team'
^researcher : 'Academic Researcher'
^reviewer : Reviewer

«private import»

«constraint def»

Candidate Eligibility Rule

constraints

eligible == ('G-01 satisfied' and 'G-02 satisfied' and 'G-03 satisfied' and 'G-04 satisfied' and 'mandatory system requirements satisfied')

features

G-01 satisfied : Boolean
G-02 satisfied : Boolean
G-03 satisfied : Boolean
G-04 satisfied : Boolean
mandatory system requirements satisfied : Boolean
eligible : Boolean

«requirement»

<G-02> Reproduce Reference Analyses

doc

Every reference analysis designated for framework evaluation can be executed from its controlled baseline and reproduces its reported result within the declared tolerance.
Evidence basis: rostamnia2026digitaltwin Secs. 8.4-8.6; rojasramirez2022comparative Sec. 6.2.

subject

enterprise : 'Academic Nanosatellite EPS Development Enterprise'

require constraints

docEvery designated reference analysis reproduces within its declared tolerance.
statement { }

«#Refinement»

«requirement»

<G-01> Complete and Valid Evidence Chains

doc

Every EPS design decision designated for framework evaluation has a complete and valid evidence chain from its source need or requirement through the applicable design elements, assumptions and input configuration, analysis result, and verification criterion. Each designated requirement set, tailored EPS model, and analysis result produced by an external engineering capability enters that chain only with its producer, source revision, controlled evidence baseline, provenance, and applicable trace relations identified and preserved.
Evidence basis: buruso2026modelbased Sec. 5.1 and Table 5; vizitiu2025optimizing Sec. 4.7 and Table 3; rostamnia2026digitaltwin Table 13.

subject

enterprise : 'Academic Nanosatellite EPS Development Enterprise'

require constraints

docEvery designated decision and accepted external artifact has a complete and valid evidence chain.
statement { }

«#Refinement»